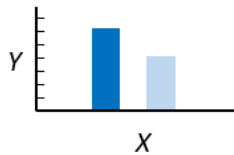


Lecture 13. Analysing Categorical Data with chi squared (χ^2)

Types of tests we are covering this year:

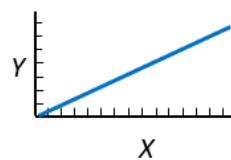


X – categorical

Y – continuous

=> **t-test**: Compares 2 groups (X)
on some measure (Y)

=> **ANOVA**: Compares 3 groups or
more



X – continuous

Y – continuous

=> **correlation**: Tests for an association
between two measures

=> **regression**: Builds a model
predicting Y based on X

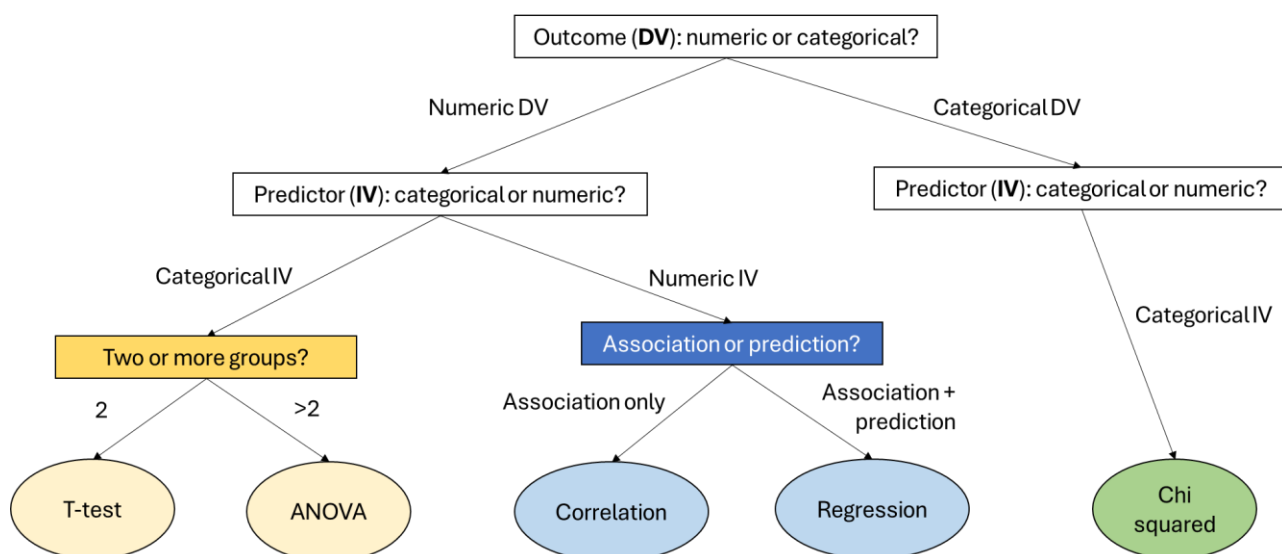
	X	
Y	31	19
	24	26

X – categorical

Y – categorical

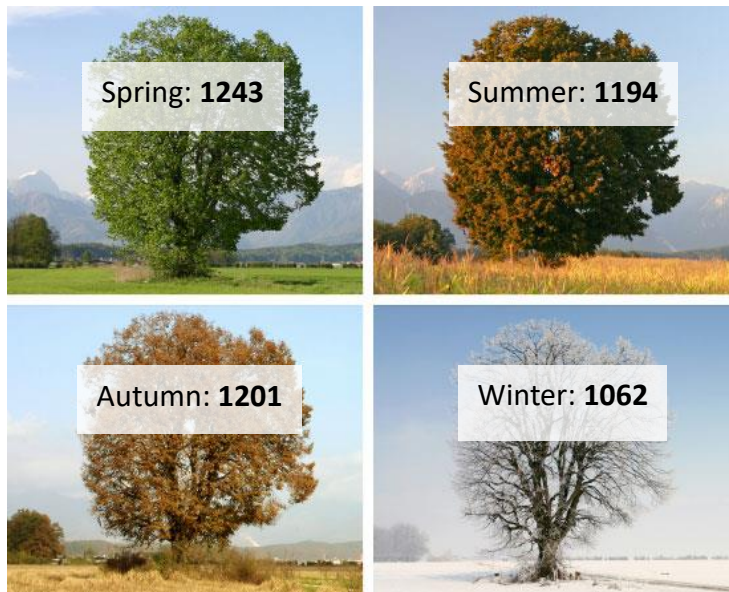
=> **χ^2 test**: Compares frequencies
(number of observations) across
different categories

- Categorical variables have values that describe a characteristic of an observation, like 'what type' or 'which category.'
- Categorical variables can be further characterised according to whether or not they have a natural order:
 - nominal, e.g. marital status (married, single, divorced, widowed)
 - ordinal, e.g. burns (first, second, third degree)



χ^2 as a 'goodness of fit' test

- Does the observed data 'fit' our expectations if the null hypothesis were true?
- Example research question: Are births equally distributed across seasons?
- Data (number of births per season):



⇒ What is the likelihood for the **observed** count of births for each season, if equal proportions of births are **expected**?

Hypotheses and expected frequencies

- If births occurred with equal frequency, the probability of births in each season would be 0.25 (1/4)
- To obtain the expected count frequencies for each season, we divide the total number of births by the number of seasons:

$$4700 \div 4 = 1175$$

or

$$4700 \times 0.25 = 1175$$

- H_0 = births occur with equal frequency across seasons
 - $P = (.25, .25, .25, .25)$
- H_1 = births do not occur with equal frequency across seasons
 - $P \neq (.25, .25, .25, .25)$

The **χ^2 statistic** (χ^2 , pronounced "kai squared") expresses the difference between Observed and Expected frequencies:

$$\chi^2 = \sum_i \frac{(\text{observed}_i - \text{expected}_i)^2}{\text{expected}_i}$$

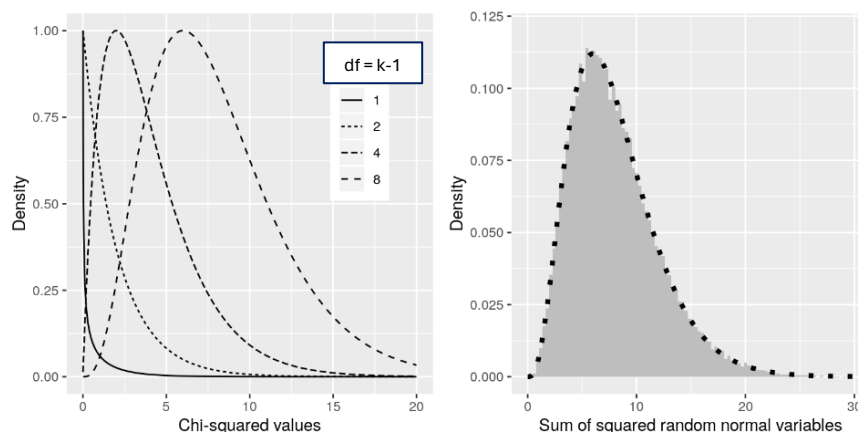
Calculating χ^2

Category	Observed frequency	Expected frequency	Difference	Difference ²	Difference ² /Expected freq
Spring	1243	1175	68	4624	3.94
Summer	1194	1175	19	361	0.31
Autumn	1201	1175	26	676	0.58
Winter	1062	1175	-113	12769	10.87
					$\Sigma = 15.7$

χ^2

χ^2 distribution for the null hypothesis

- **Normal variables** = the expected frequencies that would be observed if there were no association between the categorical variables being studied.
- Sum of squared differences that are normally distributed yield a χ^2 distribution
- χ^2 is always positive
- df is the number of k groups/categories being added together minus 1

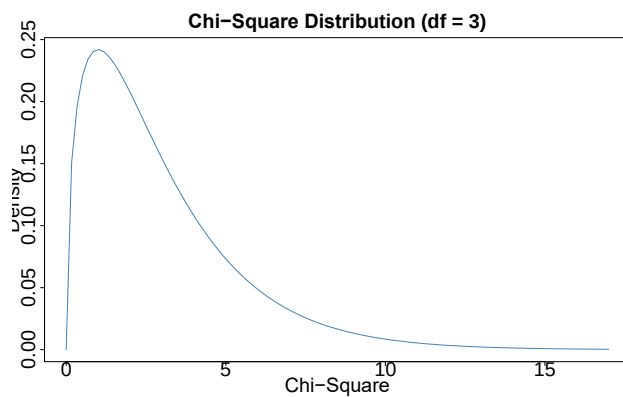


Finding the critical χ^2 value – anything larger than this will be too unlikely to result from sampling error, therefore we can reject H_0 :

```
{r}
qchisq(p = .95, df = 3)
[1] 7.814728
```

➔ For the χ^2 -distribution with 3 degrees of freedom and an alpha level of $p = .05$, the critical χ^2 -value is **7.81**.

Our χ^2 -value $15.7 >$ critical value 7.81 , so we can reject the null hypothesis.



What is the exact probability of obtaining a $\chi^2 = 15.7$ or greater if the null hypothesis was true?

```
{r}  
pchisq(q = 15.7, df = 3, lower.tail = FALSE )  
[1] 0.001306416
```

Reporting the χ^2 goodness-of-fit test

Out of 4700 births at St. Margaret's Hospital in 2021, 1243 took place in Spring, 1194 in the Summer, 1201 in the Autumn, and 1062 in the Winter season. A chi-square goodness of fit test was conducted to test whether the birth probabilities were identical across all four seasons. The results were significant, $\chi^2(3, N=4700)=15.7$, $p<.001$, suggesting that births did not occur with equal frequency across seasons.

The χ^2 test of independence (or association)

The **test of independence** is a more complex version of the chi squared test which quantifying the relationship between **two** categorical variables.

Example: Do Liberals and Conservatives differ in their preference of pet (cats vs. dogs)?

- H_0 : No association between political views and pet choice
- H_a : There is an association between political views and pet choice

<i>Observed freqs</i>	Cat	Dog
Liberals	31	14
Conservatives	16	39

To calculate the Expected frequencies (if the null hypothesis is true), we need to make a contingency table calculating the total number of Liberals, Conservatives, Cat owners, and Dog owners:

<i>Observed freqs</i>	Cat	Dog	TOTAL
Liberals	31	14	45
Conservatives	16	39	55
TOTAL	47	53	100

The expected frequency for each cell is (the *Total for its row* $\times$ the *Total for its column*) $\div$ the grand total (100):

<i>Expected freqs</i>	Cat	Dog	TOTAL
Liberals	$(45 \times 47) \div 100$ = 21.15	$(45 \times 53) \div 100$ = 23.85	45
Conservatives	$(55 \times 47) \div 100$ = 25.85	$(55 \times 53) \div 100$ = 29.15	55
TOTAL	47	53	100

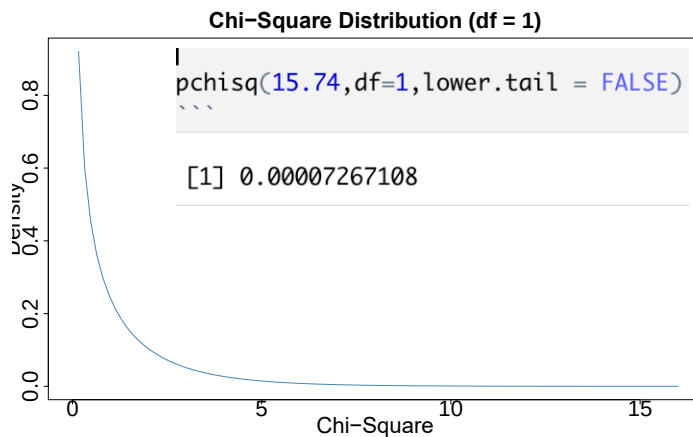
The χ^2 value is calculated in the same way as in the goodness-of-fit test:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(\text{observed}_{ij} - \text{expected}_{ij})^2}{\text{expected}_{ij}}$$

$$\chi^2 = \frac{(31 - 21.15)^2}{21.15} + \frac{(14 - 23.85)^2}{23.85} + \frac{(16 - 25.85)^2}{25.85} + \frac{(39 - 29.15)^2}{29.15} = 15.74$$

Finding the degrees of freedom: $df = (N_{rows} - 1) \times (N_{columns} - 1) = 1$

Calculating the p-value for $\chi^2=15.74$ with 1 DF:



Performing a χ^2 test in R

```
##{r}
results <- chisq.test(x = data %>% pull(PoliticalView), # 1st grouping variable
                     y = data %>% pull(PetChoice)) # 2nd grouping variable
results
##{}
```

Pearson's Chi-squared test with Yates' continuity correction

data: data %>% pull(PoliticalView) and data %>% pull(PetChoice)
X-squared = 14.18, df = 1, p-value = 0.0001661

Notice that `chisq()` by default runs the test with a continuity correction, which is used when the sample size is too small (when any of the expected frequencies in any of the cells of the contingency table is 5 or less – but this isn't a hard rule). To run the test without correction, add `correct = FALSE` to the formula:

```
##{r}
results <- chisq.test(x = data %>% pull(PoliticalView), # 1st grouping variable
                     y = data %>% pull(PetChoice), # 2nd grouping variable
                     correct = FALSE) # whether to apply continuity correction
results
##{}
```

Pearson's Chi-squared test

data: data %>% pull(PoliticalView) and data %>% pull(PetChoice)
X-squared = 15.737, df = 1, p-value = 0.00007278

Effect size: Phi (ϕ) and Cramer's V

For a 2×2 contingency table, use **Phi ϕ**

$$\phi = \sqrt{\frac{\chi^2}{N}} = \sqrt{\frac{14.18}{100}} = 0.38$$

For contingency tables bigger than 2×2, use Cramér's V, where k= field with fewest number of categories:

$$\phi = \sqrt{\frac{\chi^2}{N \times (k - 1)}}$$

Reporting the chi-square test of independence

A χ^2 test of independence revealed a significant association between political views and pet choices, $\chi^2(1, N = 100) = 14.18, p < .001, \phi = .38$. Individuals with liberal political views were more likely to choose cats than dogs as pets while individuals with conservative views were more likely to choose dogs than cats.

Assumptions of the test:

- The data in the cells should be frequencies, or **counts** of cases
- The levels (or categories) of the variables are mutually exclusive.
- **Each subject may contribute data to one and only one** cell.
- The study groups must be independent and not related.
- There are 2 variables, and both are measured as categories.
- The expected cell frequencies should be greater than 5.